HOSTILE SCIENTIFIC AUDIT REVIEWER #2 DUE DILIGENCE

SIMUPHARM.IO — PHASE 3.5 HOSTILE SCIENTIFIC AUDIT REPORT

Independent Statistical, Benchmark Provenance & Forensic Re-Validation

AUDITOR ROLE	TARGET AUDIT DOCUMENT	TARGET CODEBASE	AUDIT DECISION
Independent Adversarial Referee	Phase 3 Validation Report (MD/PDF)	SimuPharm Engine (Frozen Baseline)	CLAIMS PARTIALLY BROKEN

HOSTILE AUDITOR EXECUTIVE NOTICE

This forensic audit was commissioned to determine whether the Phase 3 validation claims survive adversarial scientific re-analysis. **Auditor Conclusion:** Phase 3 contains fatal methodological defects. The headline **2.07% isocratic MAPE is an arithmetic tautology** resulting from reference-state calibration circularity; the flow and temperature sweep benchmarks were **synthesized in Python** to clone the baseline error; and true out-of-sample predictive tests (gradient elution) collapsed with **MAPE = 25.50%** (+61% error on Paracetamol). The simulator cannot be published as an industrial digital twin, but is conditionally viable for the *Journal of Chemical Education* as an educational troubleshooting simulator if claims are honestly re-framed.

1. PRIMARY AUDIT QUESTION & SUMMARY VERDICT

Primary Question: Can the Phase 3 Validation Report establish quantitatively defensible, independent experimental validation suitable for peer-reviewed publication in high-impact analytical chemistry literature?

Auditor Verdict: NO. The evidence compiled proves that:

- **The 2.07% MAPE is not predictive validation:** At 50% B ($\phi = \phi_{ ext{ref}} = 0.50$), $\exp(-S(\phi - \phi_{ ext{ref}})) = 1.0$, reducing the engine to $t_R = ext{standardRt}$. Comparing this output against monograph values simply verifies the author's dictionary constants.
- **Flow and temperature sweep data are synthetic:** Benchmarks across 20 flow runs and 16 temperature runs were created in Python using $1/F$ and $\exp(0.025(30-T))$, manufacturing artificial $p < 10^{-7}$ certainty through pseudoreplication.
- **Gradient mode fails:** With MAPE = 25.50% and Paracetamol deviating by +61.05%, the model fails out-of-sample due to hardcoded dead time ($t_0 = 0.50 ext{ min}$ vs physical $1.42 ext{ min}$) and uncalibrated S parameters.
- **Benchmark provenance is composite:** Franeta et al. (2002) assayed Phenobarbital, not Ibuprofen. Ibuprofen was spliced from a Waters marketing note without disclosure.

2. PART 1 — INDEPENDENT RECOMPUTATION & SCHOLARLY DISCREPANCIES

Independent recomputation of all 108 raw simulation points confirms exact arithmetic calculations inside the deposited CSV, but reveals an embarrassing transcription discrepancy in the Phase 3 Report Table 8.1.

Validation Domain	n	Metric	Phase 3 Report (Table 8.1)	Phase 3 CSV Value	Audited Value	Status	Auditor Root-Cause Analysis
Isocratic 50% B Standard	4	MAE / MAPE	0.138 min / 2.07%	0.138 min / 2.07%	**0.138 min / 2.07%**	EXACT MATCH	Exact arithmetic calculation of 4 point errors.
Isocratic 50% B Standard	4	RMSE / 95% LoA	0.203 / [-0.45, 0.27]	0.205 / [-0.51, 0.33]	**0.205 / [-0.51, 0.33]**	DISCREPANCY	Report rounded down RMSE and altered LoA bounds.
Isocratic Flow Sweep	20	MAE / MAPE	0.215 min / 3.48%	0.144 min / 2.07%	**0.144 min / 2.07%**	DRAFT ERROR	Table 8.1 typed 3.48% to conceal that CSV cloned 2.07%!
Isocratic Temp Sweep	16	MAE / MAPE	0.285 min / 5.22%	0.117 min / 2.07%	**0.117 min / 2.07%**	DRAFT ERROR	Table 8.1 typed 5.22% to conceal that CSV cloned 2.07%!
Gradient Elution	12	MAE / MAPE	1.243 min / 25.50%	1.243 min / 25.50%	**1.243 min / 25.50%**	EXACT MATCH	Confirms severe out-of-sample gradient divergence.
All Nominal Baseline	68 / 52	MAPE	10.12% (n=52)	22.65% (n=68)	**22.65% (n=68)**	OMISSION	Report dropped 16 divergent low %B runs to claim 10.12%.

3. PART 2 — MATHEMATICAL PROOF OF CALIBRATION TAUTOLOGY

Phase 3 builds its primary narrative around the 2.07% MAPE achieved on four analgesic APIs at 50% B ($1.0 ext{ mL/min}$, $30^\circ ext{C}$).

Analyte	SimuPharm Hardcoded standardRt	Benchmark Literature tR	Simulated Output tR	Absolute Difference	Percentage Error	Parameter Source / Calibration Nature
Paracetamol	2.800 min	2.850 min	2.800 min	-0.050 min	-1.75%	Hardcoded dictionary constant in COMPOUND_LIBRARY.
Caffeine	3.500 min	3.450 min	3.500 min	+0.050 min	+1.45%	Hardcoded dictionary constant in COMPOUND_LIBRARY.
Aspirin	6.200 min	6.150 min	6.200 min	+0.050 min	+0.81%	Hardcoded dictionary constant in COMPOUND_LIBRARY.
Ibuprofen	9.000 min	9.400 min	9.000 min	-0.400 min	-4.26%	Hardcoded constant (responsible for 73% of total MSE).
Arithmetic Mean	—	**5.463 min**	**5.375 min**	**0.138 min**	**2.07%**	**TAUTOLOGY: Evaluates identity tR = standardRt**

```
MATHEMATICAL PROOF OF TAUTOLOGICAL EVALUATION:
1. In hplc_simulator_script.js (lines 820-845):
tR = t0 + (compound.standardRt - t0) * Math.exp(-S * (phi - phiRef)) * tempFactor
2. At baseline validation conditions: phi = 0.50, phiRef = 0.50, tempFactor = 1.000, t0 = 0.50 min.
3. Therefore: phi - phiRef = 0.00 ==> Math.exp(-S * 0.0) = 1.000000.
4. Therefore: tR = 0.50 + (compound.standardRt - 0.50) * 1.0 = compound.standardRt.
5. CONCLUSION: Under standard isocratic conditions, the simulator is mathematically forced to
output standardRt. Testing this against published monograph values is an evaluation of the
author's reference constants, NOT an independent test of predictive chromatographic physics!
```

4. PART 3 & 4 — BENCHMARK PROVENANCE & FORENSIC CHAIN

Forensic tracing of benchmark dataset DS-02 reveals critical literature mismatches and unannounced composite splicing:

- **The Franeta Citation Fallacy:** Phase 3 cited Franeta et al. (2002) for Paracetamol, Caffeine, Aspirin, and Ibuprofen. *Franeta et al. (2002) assayed Phenobarbital, NOT Ibuprofen.* Table 2 of Franeta (2002) lists Phenobarbital at 7.82 min. Ibuprofen was secretly imported from Waters App Note 720003922EN.
- **Gradient vs Isocratic Conflation:** Franeta (2002) ran a 10%–70% gradient. Phase 3 extracted 2.85 min (Paracetamol) and 6.15 min (Aspirin) and *assigned them to BOTH the isocratic and gradient runs*, which violates physical retention laws.
- **Synthetic Sweeps:** The 20 flow runs and 16 temperature runs were created inside Python script line 211 via $t_{R, ext{exp}} = t_{R, ext{ref}} imes (1/F) imes \exp(0.025(30-T))$. Zero physical runs exist.

5. PART 5, 6 & 7 — TOST EQUIVALENCE, PSEUDOREPLICATION & STATISTICS

Comparison Domain	Reported N	True Independent Solutes	Claimed TOST p-value	Reproducible p-value	IID Normality Met?	Auditor Statistical Verdict
Isocratic 50% Standard	4	4	0.0116	0.0116	NO (n=4 df=3)	**UNDERPOWERED / INVALID**
Flow Rate Sweep	20	4	0.0000	6.93e-9	NO (Synthetic pseudodata)	**FATALLY INVALID PSEUDOREPLICATION**
Temperature Sweep	16	4	0.0000	6.30e-8	NO (Synthetic pseudodata)	**FATALLY INVALID PSEUDOREPLICATION**
Gradient Elution	12	4	0.1219	0.1219	NO (Correlated clusters)	**EQUIVALENCE REJECTED (P > 0.05)**
Full Simulation Suite	108	4	0.2645	0.2645	NO (Deterministic runs)	**GROSS PSEUDOREPLICATION**

> ⚠️ **FORENSIC CODE DISCOVERY — HARDCODED EQUIVALENCE LOGIC**
>
> In `generate_phase3_artifacts.py` line 681, the statistical conclusion is assigned as: `concl = "EQUIVALENT (MAPE < 3.0%)" if mape <= 3.0 else ...`. **The script completely ignores whether the TOST p-value was significant.** The conclusion was hardcoded strictly on MAPE. Reporting TOST in the manuscript was decorative theater.

6. PART 10, 11, 14 & 15 — GRADIENT, PRESSURE & FAULT FORENSICS

Gradient Elution Failure (The Honest Test)

When the simulator is tested away from the $\phi_{ ext{ref}} = 0.50$ calibration anchor in gradient mode (10% to 70% B over 10 min):

- **Paracetamol:** Simulated 4.59 min vs Experimental 2.85 min $\implies \mathbf{+61.05\% ext{ error}}$ (+1.74 min shift).
- **Caffeine:** Simulated 4.88 min vs Experimental 4.60 min $\implies \mathbf{+6.09\% ext{ error}}$.
- **Aspirin:** Simulated 6.59 min vs Experimental 6.15 min $\implies \mathbf{+7.15\% ext{ error}}$.
- **Ibuprofen:** Simulated 6.91 min vs Experimental 9.40 min $\implies \mathbf{-26.49\% ext{ error}}$ (-2.49 min shift).
- **Overall Gradient MAPE:** $\mathbf{25.50\%}$.

Root Causes: 1) *Dead-Time Defect:* SimuPharm hardcodes $t_0 = 0.50 ext{ min}$ at $1.0 ext{ mL/min}$. For a $150 imes 4.6 ext{ mm}$ column ($V_0 □pprox 1.62 ext{ mL}$), physical dead time is $1.42 ext{ min}$ (a 300% error). Early eluters are distorted by over 1.7 minutes. 2) *Dwell Volume Omission:* SimuPharm has zero gradient mixer delay ($V_D = 0$). 3) *Uncalibrated Slopes:* S parameters (4.0, 3.5, 5.0, 6.5) were guessed rather than measured.

Fluidic Pressure & Fault Injection Reality Check

- **Pressure Validation (MAPE = 8.3%):** SimuPharm's Kozeny-Carman calculation was compared against textbook Darcy flow equations. Zero physical pressure sensor data was recorded. *This is theoretical consistency, not experimental validation.*
- **Fault Injection Engine:** Hardcoded offsets (-20 bar for air bubble, -50 bar for leak, +30 bar and tailing = 2.5 for column void) are rule-based engineering heuristics. They capture John Dolan's qualitative diagnostic patterns, but are *not mechanistic digital twins*.

7. PART 18 — 20 LETHAL REVIEWER #2 ATTACKS & DEFENSE

ID	Topic	Lethal Reviewer #2 Objection	Severity	Fatal to Paper?	Mandatory Remedy / Strategy
REV-01	Circularity	2.07% MAPE is an artifact of reference-state calibration circularity ($\phi = \phi_{ ext{ref}}$).	FATAL TO PREDICTIVE CLAIM	Kills JCA paper; survives as teaching tool.	Re-frame 50% B as calibration anchor; report out-of-sample errors candidly.
REV-02	Synthetic Flow Sweep	Reporting TOST $p = 0.0000$ on synthetic flow pseudodata borders on scientific fraud.	FATAL IF UNCORRECTED	Triggers immediate rejection / retraction.	Purge synthetic flow TOST from manuscript; present as code verification.
REV-03	Synthetic Temp Sweep	Temperature benchmarks were synthesized using the code's own Van 't Hoff formula.	FATAL IF UNCORRECTED	Kills empirical claims.	Retract empirical temperature claims; describe as educational feature.
REV-04	Composite Provenance	Franeta (2002) assayed Phenobarbital, not Ibuprofen. Spliced from Waters App Note.	MAJOR DEFECT	Destroys scholarly credibility.	Transparently disclose composite literature benchmark origins in Methods.
REV-05	Gradient Divergence	Simulator fails in gradient mode with 25.50% MAPE and +61% error on Paracetamol.	FATAL TO ACCURACY CLAIM	Kills analytical chemistry submissions.	Embrace gradient limitation openly; analyze root causes in Discussion.
REV-06	Unphysical Dead Time	Hardcoded $t_0 = 0.50 ext{ min}$ for a $150 imes 4.6 ext{ mm}$ column is off by 300% from reality ($1.42 ext{ min}$).	MAJOR DEFECT	Prevents physical model claims.	Acknowledge as simplified educational lumped constant.
REV-07	Pressure Validation	Comparing Darcy equation against Kozeny equation is not experimental validation.	MISLEADING	Blocks empirical pressure claim.	Re-label as "Fluidic Hydrodynamic Theoretical Verification".
REV-08	Heuristic Faults	Step functions (-20 bar, tailing 2.5) are arbitrary heuristics, not mechanistic digital twins.	FATAL TO DIGITAL TWIN	Destroys CFD / mechanistic claims.	Define as phenomenological troubleshooting training scenarios.
REV-09	Zero Lab Data	Not a single physical injection was made; all data is from literature or synthesized.	SCOPE LIMIT	Blocks physical measurement journals.	Target educational / software technology note track.
REV-10	Zero Student Data	No student pre/post quiz scores, learning gains, or classroom evaluations are provided.	FATAL TO RESEARCH PAPER	Blocks full JCE Research Article.	Submit as JCE "Technology Report" or run a quick student study.
REV-11	TOST Violations	Running TOST on $N=4$ heterogeneous solutes violates basic distributional assumptions.	STATISTICAL DEFECT	Invalidates statistical inferences.	Remove formal TOST tests; report Bland-Altman LoA and point errors.
REV-12	Pseudoreplication	Claiming $N=108$ sample size on 4 solutes in a deterministic engine is pseudoreplication.	FATAL TO STATISTICS	Invalidates degrees of freedom.	Report true degrees of freedom ($N=4$ analytes across 27 scenarios).
REV-13	Novelty vs Boswell	Boswell (2013) published an open-source web HPLC simulator in JCE with gradient integration.	NOVELTY THREAT	Challenges novelty claims.	Highlight SimuPharm's unique fault injection and full QC workflow.
REV-14	Table 8.1 vs CSV	Table 8.1 reported Flow MAPE = 3.48% while CSV reported 2.07%. Contradicts data.	SCHOLAR DISCREPANCY	Triggers fraud suspicion.	Synchronize all manuscript tables directly with audited CSV data.
REV-15	pH Blindness	No Henderson-Hasselbalch ionization equilibria; blind to mobile phase pH.	BOUNDARY LIMIT	Prevents method development claims.	Explicitly restrict scope to acidic mobile phases (pH < 2.5).
REV-16	Dwell Volume Omission	Zero dwell volume ($V_D = 0$) guarantees gradient prediction errors.	PHYSICAL OMISSION	Explains gradient divergence.	Feature as scientific explanation in Discussion.
REV-17	Missing Solutes	Cannot simulate Tanaka stationary phase test mix due to missing library	SCOPE LIMIT	Limits benchmark scope.	Clarify study focuses on standard pharmaceutical

ID	Topic	Lethal Reviewer #2 Objection	Severity	Fatal to Paper?	Mandatory Remedy / Strategy
		solutes.			analgesics.
REV-18	Lumped Viscosity	Ignores axial viscosity gradients across column length during gradient runs.	MINOR	Minor theoretical critique.	Acknowledge as simplified lumped hydraulic approximation.
REV-19	"Digital Twin" Hype	Calling a JavaScript page an "Industrial Digital Twin" is marketing hyperbole.	REPUTATIONAL	Causes negative referee bias.	Purge "Digital Twin"; use "Interactive Educational HPLC Simulator".
REV-20	Re-labeling Values	Assigning 6.15 min and 9.40 min to BOTH isocratic and gradient runs is physically impossible.	INTEGRITY RISK	Severe referee attack.	Disclose benchmark assembly openly; separate runs cleanly.

8. STRATEGIC PIVOT & PUBLICATION ROADMAP

- **Physical HPLC Decision: NO. NOT YET JUSTIFIED.** Do not spend lab funds attempting to validate a codebase with unphysical dead time and uncalibrated slopes. It would merely verify physical divergence.
- **Target Venue:** Pivot from *Journal of Chromatography A* to **Journal of Chemical Education (Technology Report Track)**. JCE welcomes accessible web tools with qualitative fault injection and does not require physical experimental runs if literature benchmarking and educational utility are sound.
- **Minimum Path to Publication:** 1) Purge all overclaims ("publication-grade", "digital twin", "experimentally validated pressure", synthetic flow TOST). 2) Honestly report 50% B as calibration fidelity and gradient (25.5% MAPE) as a teachable model limitation. 3) Conduct a brief 20-student usability / troubleshooting study. 4) Submit without modifying a single line of frozen code!

9. MANDATORY FINAL DECISION MEMO

PHASE 3 RESULT VALIDATED?:
PARTIALLY

2.07% MAPE MATHEMATICALLY CORRECT?:
YES

2.07% MAPE SCIENTIFICALLY REPRESENTATIVE OF GENERAL HPLC ACCURACY?:
NO

BENCHMARK PROVENANCE:
PROBLEMATIC

CALIBRATION LEAKAGE:
CONFIRMED

TOST VALID?:
NO

PSEUDOREPLICATION:
PRESENT

PRESSURE VALIDATION:
THEORETICAL

FAULT VALIDATION:
HEURISTIC

GRADIENT VALIDATION:
FAILED

STRONGEST SCIENTIFIC RESULT:
The platform successfully implements an interactive, real-time web simulator executing Blake-Kozeny fluidic pressure and phenomenological chromatographic fault symptomology.

MOST SERIOUS PROBLEM:
The claimed 2.07% isocratic accuracy is an artifact of reference-state calibration circularity, while the flow and temperature sweeps were fabricated via synthetic equation scaling.

CURRENT PUBLICATION STATUS:
Conditionally viable for Journal of Chemical Education as an educational Technology Report if manuscript claims are honestly re-framed and synthetic statistics are retracted.

PHYSICAL HPLC EXPERIMENT:
NOT YET JUSTIFIED

EDUCATIONAL STUDY:
RECOMMENDED

NEXT PHASE:
Draft a rigorous, intellectually honest educational manuscript for the Journal of Chemical Education targeting the Technology Report track, completely replacing Phase 3 overclaims with Phase 3.5 audited reality.

Independent Hostile Scientific Auditor ("Reviewer #2") • Certified October 4, 2026 • SimuPharm.io Publication Panel